

4(a)	<i>idea of shape needs to be in the answer for mp2 and mp4 I 'substrate changes shape'</i> <i>any three from</i> 1 active site is not (fully) complementary to substrate ; 2 active site, changes shape / moulds around, to fit the substrate ; A conformational change for shape change 3 <u>enzyme-substrate complex / ESC</u>, forms ; 4 active site returns to original shape on release of product ; 5 AVP ; e.g. change of shape (to give complementary fit) lowers activation energy / puts strain on bonds / AW <i>ref. to binding site / catalytic site (of active site)</i>	3
4(b)(i)	A <i>peroxidase for enzyme</i> <i>any three from</i> <i>rate of reaction increases as substrate concentration increases to 0.33 – 0.35 mmol dm⁻³</i> 1 (up to 0.33 to 0.35 mmol dm⁻³) some active sites are not occupied ; 2 more collisions between, enzyme / active site, and substrate molecules ; A more collisions lead to increase in formation of enzyme-substrate complexes <i>rate of reaction remains constant above 0.33–0.35 mmol dm⁻³</i> 3 all active sites, occupied / saturated ; 4 correct reference to limiting factor for slope ; A limits / limiting e.g. at low substrate concentrations, substrate concentration limiting / enzyme concentration not limiting 5 correct reference to limiting factor for plateau ; A limits / limiting e.g. at high substrate concentrations, enzyme concentration limiting / substrate concentration not limiting	3
4(b)(ii)	<i>idea that determined V_{max} / maximum rate / ref. to 5.6 (μmol min⁻¹) ;</i> A if V_{max} shown on graph K_m is the substrate concentration at half the V_{max} ;	2

Question	Answer	Marks
4(c)	<i>any three from</i> <i>in respiring tissue</i> 1 (carbonic anhydrase) catalyses the reaction between water and carbon dioxide in, red (blood) cells / erythrocytes ; 2 to form carbonic acid, which dissociates to form HCO_3^- and H^+ ; A from equation A use of word ion(s) 3 HCO_3^-, moves / passes / diffuses (through membrane proteins from red blood cells) into plasma ; 4 (activity of enzyme) maintains (steep) concentration gradient for diffusion of carbon dioxide into red blood cells ; 5 <i>idea that</i> so helps to remove large quantities of carbon dioxide ; <i>in lungs</i> 6 (carbonic anhydrase) converts HCO_3^- back into carbon dioxide so can be, excreted / removed / exhaled ; <i>if described earlier A</i> catalyses reverse reaction so carbon dioxide can be, excreted / AW	3

Question	Answer	Marks
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